

PUBLISHED

Question	Answer	Marks	Guidance
6(a)	$H_0: \mu_B = \mu_A \quad H_1: \mu_B > \mu_A$	B1	If in words, must contain ‘ population means’. Allow $H_0: \mu_d = 0$, $H_1: \mu_d > 0$ if defined or consistent with working.
	Differences: 5, 0, –1, 6, 1, –2, 8, 7, 15	M1	Attempt at signed differences.
	$s_d^2 = \frac{1}{8} \left('405' - \frac{'39'^2}{9} \right) \quad [= 29.5]$	M1	Correct form for s_d^2 , note that $s_d = 5.43$.
	$\frac{\frac{'39'}{9}}{\sqrt{\frac{'29.5'}{9}}} = 2.393$	M1	Correct form.
		A1	AWRT 2.39.
	'2.393' > 1.860, reject H_0 / significant.	M1	Compare <i>their</i> 2.393 with 1.860 and appropriate result (may be in terms of H_1). Allow M1 for comparison from 2-sample test using 1.746. Test result may be implied by an attempt at an appropriate conclusion in context ignoring hypotheses.
	Sufficient evidence to suggest new training programme results in reduced times. OR Sufficient evidence to support the coach’s belief.	A1	Correct conclusion from correct working ignoring hypotheses. In context. Level of uncertainty in language is used (for example, not ‘prove’). Do not accept statements such as “there is insufficient evidence to suggest...”.
		7	
6(b)	The population of differences may not be normally distributed. The population of differences may not be symmetrical (and hence not normal).	B1	Must mention population/distribution and differences. Accept “the sample may not be random” or “times may not be independent”.
		1	
6(c)	A Wilcoxon matched- pairs signed-rank test. OR A paired -sample sign test.	B1	Must refer to pairs / paired. Accept “ paired Wilcoxon test” or “ paired sign test”.
		1	